



Capstone Project Report

1. Introduction

This capstone project demonstrates a complete SOC workflow including attack simulation, detection, triage, containment, escalation, and reporting.

The simulation involved exploiting a vulnerable system using **Rapid7 Metasploit**, detecting malicious activity using **Wazuh Inc. Wazuh**, performing response actions using **CrowdSec CrowdSec**, incident escalation through **StrangeBee TheHive**, and documentation using **Google LLC Google Docs**.

The project follows industry-standard incident response practices and aligns with **MITRE Corporation ATT&CK methodology** and **SANS Institute incident response guidelines**.

2. Objective

The objective of this project is to simulate a real-world cyber attack and perform a complete security incident response lifecycle.

Objectives

- Simulate an attack on a vulnerable system.
 - Detect malicious activity using monitoring tools.
 - Perform alert triage and investigation.
 - Contain and mitigate the threat.
 - Escalate the incident to higher response level.
 - Document and report the incident.
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3. Tools Used

- Metasploit Framework - Attack simulation and exploitation
 - Wazuh - Security monitoring and alert detection
 - CrowdSec - Threat blocking and containment
 - TheHive - Incident case management and escalation
 - Google Docs - Incident documentation
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4. Methodology / Implementation

4.1 Attack Simulation

A vulnerable **Metasploitable2** virtual machine was used as the target system. The Samba service vulnerability was exploited using the Metasploit module:

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exploit/multi/samba/usermap_script
```

Attack Steps

- Identified open Samba service.
- Executed exploitation module.
- Gained unauthorized shell access.
- Generated malicious system activity.

This simulated a remote exploitation scenario commonly observed in enterprise environments.

4.2 Detection and Alert Triage

Wazuh was configured to monitor system logs and detect exploitation behavior. The attack generated alerts indicating suspicious network activity.

Detected Alert Details

Timestamp	Source IP	Alert Description	MITRE Technique
2026-02-20 13:00:00	192.168.110.140	Samba exploit detected	T1210 – Exploitation of Remote Services

Triage Actions

- Verified alert authenticity.
- Checked system logs.
- Assessed attack severity.
- Classified incident as high priority.

The alert was confirmed as a true positive security incident.

4.3 Response and Containment

Immediate containment actions were performed to prevent further compromise.



Containment Steps

- Isolated compromised virtual machine.
- Blocked attacker IP using CrowdSec firewall rules.
- Terminated unauthorized session.
- Verified containment using ping test.

Verification

- Blocked IP was unable to communicate with target system.
- System returned to secure state.

4.4 Incident Escalation

The incident was escalated to Tier 2 security analysts using TheHive case management system.

Case Summary

A high-severity security incident was detected involving remote exploitation of a Samba service on a Metasploitable2 virtual machine. The attack originated from IP address 192.168.1.101 and was identified through Wazuh monitoring alerts. Initial triage confirmed unauthorized system access through exploitation of a known vulnerability. Immediate containment actions were performed by isolating the affected machine and blocking the attacker's IP using CrowdSec. Evidence including logs and network activity was preserved for investigation. Due to the critical nature of the exploit and potential system compromise, the case has been escalated to Tier 2 analysts for further forensic analysis and remediation planning.

4.5 Evidence Collection

Digital evidence was collected for forensic investigation.

Collected Evidence

- System logs from compromised machine.
- Network traffic records.
- Alert records from Wazuh.
- Incident documentation.

Evidence integrity was maintained using proper documentation procedures.



5. Incident Response Report (SANS Format)

Executive Summary

A simulated cyber attack was performed to evaluate SOC detection and response capabilities. The attack involved exploitation of a Samba vulnerability on a Metasploitable2 virtual machine using Metasploit. The attack was successfully detected by Wazuh, and immediate response actions were executed to contain the threat.

Timeline

The attack was initiated at 14:00, followed by alert generation within minutes. SOC analysts verified the alert, performed triage, and confirmed unauthorized system access. The compromised system was isolated, and the attacker's IP was blocked using CrowdSec. The incident was escalated to Tier 2 for further investigation.

Recommendations

Organizations should implement continuous monitoring, regular vulnerability patching, and strict network access controls. Security teams should also maintain incident response procedures and conduct periodic simulations to improve response readiness.

6. Non-Technical Incident Briefing

A cybersecurity incident was detected involving unauthorized access to a system through a known software vulnerability. Security monitoring tools identified suspicious activity and alerted the SOC team. Immediate actions were taken to isolate the affected system and block the attacker's access, preventing further damage. The incident was successfully contained, and no critical data loss occurred. Security specialists are conducting further analysis to understand the attack and strengthen system defenses. Preventive measures, including improved monitoring and vulnerability management, will be implemented to reduce the risk of similar incidents in the future.

7. Key Findings

- Vulnerability exploitation successfully simulated.
 - Security monitoring detected malicious activity.
 - Incident response process executed effectively.
 - Rapid containment prevented further compromise.
 - SOC workflow validated in controlled environment.
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8. Conclusion

The capstone project successfully demonstrated a complete SOC workflow from attack simulation to incident response and reporting. The exercise provided practical experience in detecting, analyzing, and mitigating cybersecurity threats using industry-standard tools and methodologies. Implementing structured SOC processes enhances organizational security posture and improves incident response readiness.